

Knowledge as Dynamics: Distilling, Composing, and Inheriting the Latent Vector Fields of Neural Networks

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Abstract

An autoencoder implicitly defines a vector field on its latent space: iterating $f = E \circ D$ displaces every point by $V(z) = f(z) - z$, and training induces attractors in this field that encode what the network has learned (Fumero et al., 2025). Prior work treats this field as an analysis object. We instead treat a network’s learned knowledge *as* its dynamics, and characterize the field as a transmission channel through five laptop-scale, pre-registered experiments. We find: (1) *sufficiency*—regressing a student’s field onto a teacher’s transfers the full attractor landscape with zero training data (0.86 basin agreement against a 0.99 ceiling; output distillation 0.53, static feature matching 0.15); (2) *order*—the seed-to-seed *reliability* of transmission changes discontinuously exactly where the teacher’s field becomes fully convergent, coinciding with its memorization-to-generalization transition; (3) *resolution* (restated in v1.1)—basin-level fidelity is governed by *directional* fidelity in decision-relevant regions, near basin boundaries, not by mean field error, which anti-correlates with basin transfer across carriers of the same field; parametric students fail at basin level on MNIST (16-dimensional latents) while a lookup carrier of the identical field climbs without plateau in its sample budget, and chart compression destroys transfer by collapsing inter-basin margins (grafting through a foreign encoder’s out-of-distribution chart: 0.14 versus 0.75 under a designed frame); (4) *composition*—the fields of two disjoint specialists union into one data-free student (0.82 joint agreement), which behavioral merging cannot do at all (0.13, grafted side 0.00); and (5) *heredity with selection* (restated in v1.1)—knowledge survives five generations of data-free re-distillation essentially undegraded, and memorization-regime structure cannot enter a lineage carried by parametric re-distillation; v1.1 locates this selection in the lossy carrier, not the channel: a lossless read-out carrier holds unions and lineages at its ceiling and transmits the same memorized structure faithfully. Revision v1.1—prompted by an independent replication and re-diagnosis received within a day of release (Section 5)—adds carrier, budget, and boundary-geometry experiments: a 10^6 -pair lookup carrier reaches 0.41 basin agreement on MNIST where parametric students reach 0.09–0.19 against a 0.71 ceiling; the ceiling itself is fractal basin-boundary geometry, measured by the uncertainty exponent ($\alpha = 0.45$); and a new two-hop experiment—a parametric student supervised by a 16k-pair non-parametric reconstruction of the field—reaches 0.31–0.50 in half

*Revision v1.1 (July 10, 2026) added Section 5 and restated Laws 3 and 5 (sharpening Law 2); v1.2 (July 10, 2026) welcomes Ivan Nemytchenko as co-author. See the contribution statement and the revision note.

†Entered the project through an independent, from-scratch replication and re-diagnosis of the v1.0 release, filed as repository issue #1 within a day of publication and validated in Section 5; co-author from v1.2.

its seeds, locating the parametric bottleneck in reliability, not capacity. We release the complete harness; the full suite runs in roughly twenty-two minutes on a laptop.

1 Introduction

Knowledge distillation transfers what one network has learned into another by matching observable quantities of the teacher: output distributions (Hinton et al., 2015), intermediate features (Romero et al., 2015), pairwise relations among embeddings (Park et al., 2019; Tung and Mori, 2019; Tian et al., 2020), or input–output Jacobians (Srinivas and Fleuret, 2018). All of these targets are *static*: they describe where the teacher places points, or how its outputs respond locally to inputs, at a single application of the map.

Neural networks also possess a dynamical observable. Fumero et al. (2025) showed that any autoencoder defines a vector field on its latent space: the composite map $f(z) = E(D(z))$ displaces each latent point by $V(z) = f(z) - z$, and iterating f reveals attractors that emerge from training and characterize the network’s learned structure, including its memorization and generalization behavior. That work, and the small literature following it, uses the field to *analyze* a given model.

This paper takes the further step of treating the field as the knowledge itself, and asks what kind of *channel* it is. If what a network knows is how it moves representations under its own re-encoding, then transferring, combining, and preserving knowledge become operations on vector fields: regression (distillation), superposition under a change of coordinates (composition), and iterated re-distillation (inheritance). We study each operation empirically, at deliberately small scale, with falsification criteria fixed before each run and with negative results reported alongside positive ones.

Our contributions are five empirical regularities—stated here as laws with their demonstrated scope, and revisited as conjectures beyond it in Section 6:

1. **Sufficiency (Section 4.1).** The field alone carries the learned structure. A student trained only to match V_t at sampled latent points—no data, no weights—reproduces the teacher’s basins of attraction at 0.86 agreement (ceiling 0.99). The pre-registered kill switch does not fire: output distillation reaches 0.53, static latent matching 0.15, training from scratch on the same data 0.00.
2. **Order (Sections 4.2–4.3).** Transmissibility is governed by the source’s dynamical order. Across a sweep of teacher regimes, the *reliability* of transmission is the discontinuous quantity: seed variance collapses from ± 0.25 to ± 0.00 at exactly the noise level where the teacher’s field becomes fully convergent. A stronger hypothesis—that copyability tracks test-time robustness—is falsified by the same data. v1.1 adds a teacher-only refinement: the uncertainty exponent α of the teacher’s basin boundaries (Grebogi et al., 1983; McDonald et al., 1985) predicts the transmission ceiling where convergence fraction alone does not (Section 5).
3. **Resolution — restated in v1.1 (Sections 4.4, 4.5 and 5).** Basin-level fidelity is governed by *directional* fidelity in decision-relevant regions—near basin boundaries—together with carrier budget; mean field error is the wrong observable and anti-correlates with basin transfer across carriers of the same field. v1.0 attributed the MNIST failure (every parametric method fails at basin level, best 0.19 against a 0.71 ceiling) to a residual-compounding inequality; v1.1 overturns that diagnosis: a lookup carrier of the identical field climbs without plateau in its pair budget, and the ceiling is the teacher’s own fractal boundary geometry.

The chart-compression failure stands: grafting knowledge through a foreign encoder’s out-of-distribution chart collapses inter-basin margins (0.14, versus 0.75–0.84 when the same knowledge is grafted isometrically).

4. **Composition (Section 4.5).** Fields union. One student, distilled data-free from a gated superposition of two disjoint specialists’ fields in a *designed* common frame, holds both knowledge sets (joint agreement 0.82, ceiling 0.98). Merged output distillation has no mechanism for placing disjoint knowledge and scores 0.13, with the grafted side at exactly 0.00.
5. **Heredity with selection — restated in v1.1 (Sections 4.5 and 5).** The channel iterates. Five generations of data-free re-distillation retain 91% of the ancestor’s joint agreement with stable attractor structure—transmission is not photocopy decay. It also selects: a memorization-regime teacher’s structure never enters the SGD-carried lineage (≤ 0.24 throughout) while ordered structure rides beside it at 0.89–1.00. v1.1 locates the selection: a lossless read-only carrier holds unions and lineages at its ceiling *and transmits the memorized structure faithfully* (0.68, flat over five generations). What is heritable through a lossy parametric carrier at fixed budget is what is general; the field channel itself is nearly lossless and unselective.

Revision v1.1. Within a day of the v1.0 release, Ivan Nemytchenko—then an outside reader, a co-author from v1.2—contributed an independent, from-scratch replication and re-diagnosis of these results through the repository’s issue tracker.¹ Section 5 validates its four claims inside the original harness—three are confirmed, one partially—and reports a new experiment the exchange prompted (two-hop distillation). Laws 2, 3 and 5 above are stated in their corrected v1.1 forms; Sections 4.4 and 4.5 preserve the v1.0 experiments and mark where their original interpretation is superseded.

2 Related Work

Distillation of static structure. Classical knowledge distillation matches softened output distributions (Hinton et al., 2015); FitNets extend the target to intermediate feature activations (Romero et al., 2015). A family of methods transfers *relational* structure: Relational Knowledge Distillation matches pairwise distances and angles among embeddings (Park et al., 2019), Similarity-Preserving distillation matches batch similarity matrices (Tung and Mori, 2019), and Contrastive Representation Distillation maximizes mutual information between teacher and student representations (Tian et al., 2020). All are functions of a single forward pass. Srinivas and Fleuret (2018) match input–output Jacobians, a first-order local property of the feed-forward map; this too concerns one application of the network, not the iteration of a self-map.

Distillation of manifolds and embeddings. Recent work distills latent geometry as a static object: MEDAL (2026) fit a constrained autoencoder whose bottleneck reproduces a fixed embedding, and Distilling Latent Manifolds (2026) distill VAE encoders and observe that resolution-consistent latent manifolds transfer. Neither considers the iterated encode–decode map or its attractors.

¹<https://github.com/Veso-AI-Open-Source/knowledge-as-dynamics/issues/1>; see the contribution statement.

Latent-space dynamics. Fumero et al. (2025) establish the latent vector field of $E \circ D$ as a representation of a trained network, connect its attractors to memorization and generalization, and pose the alignment of fields across models as an open problem. Pappone et al. (2025) analyze the per-loop displacement dynamics of recurrent-depth transformers and exploit them for early exiting. Both treat dynamics as something to measure, not to transfer. Score-based and diffusion models (Ho et al., 2020; Song et al., 2021) parameterize vector fields by construction; the field studied here is instead *emergent*—induced by re-encoding rather than trained as a generative target—though the transmission machinery developed below may port to trained fields as well. Our alignment and frame constructions (Sections 3.3 and 3.5) operationalize the open problem of Fumero et al. (2025) in the service of transfer. In v1.1, the classical theory of fractal basin boundaries (Grebogi et al., 1983; McDonald et al., 1985) supplies the geometry governing transmission ceilings (Section 5), and the Gaussian-process field reconstruction examined there descends from Wiener-filter estimation of continuous fields (Hoffman and Ribak, 1991; Rasmussen and Williams, 2006).

Positioning. To the best of our knowledge, based on a systematic search of arXiv and citation indices conducted in July 2026 (including all works then citing Fumero et al., 2025), no prior publication trains a student network to match a teacher’s iterated-map displacement field, composes two models’ fields into one student, or studies the generational transmission of such fields. The methods above transfer static snapshots of the map; we transfer, combine, and propagate how representations *move* under the model’s own re-encoding.

3 The Object and the Channel

3.1 The latent vector field

Let $E : \mathcal{X} \rightarrow \mathcal{Z}$ and $D : \mathcal{Z} \rightarrow \mathcal{X}$ be the encoder and decoder of a trained autoencoder. Following Fumero et al. (2025), the composite map on latent space,

$$f(z) = E(D(z)), \quad V(z) = f(z) - z, \quad (1)$$

defines a discrete dynamical system $z_{t+1} = f(z_t)$. Attracting fixed points z^* (with $V(z^*) = 0$) and their basins constitute a compact summary of what the network has learned: standard training biases produce contraction toward the learned data manifold, and the attractor landscape encodes its structure.

3.2 Vector-field distillation

Given a frozen teacher with field V_t , we train a freshly initialized student by regressing its field onto the teacher’s at sampled latent points:

$$\mathcal{L}_{\text{VF}}(s) = \mathbb{E}_{z \sim p(z)} \|V_s(z) - V_t(z)\|_2^2, \quad (2)$$

where $p(z)$ is a sampling prior over the latent region of interest. The objective is *data-free*—no training example appears in Eq. (2); the teacher is queried only through its latent self-map—and it constrains the student’s composite map on latent space while leaving its data-space behavior unconstrained (a small reconstruction anchor is added where data-space behavior is also desired). A multi-step variant matches iterates f^k directly for $k \in K$; anticipating the results, it never outperformed the one-step objective and is reported as an ablation.

3.3 Comparing fields across models

A student’s latent chart need not coincide with the teacher’s, so fields are compared up to a change of coordinates. Given an invertible affine map $A(z) = zW + b$ from student to teacher coordinates, the student’s field in teacher coordinates is

$$\widehat{V}_s(z) = A(f_s(A^{-1}(z))) - z. \quad (3)$$

For every trained student we fit W, b by least squares between student and teacher encodings of the training data, and grant each method the better of {identity, fitted affine} by field error on a held-out probe set—a best-case protocol that is deliberately generous to the baselines. Affine maps cannot capture arbitrary nonlinear re-parameterizations; we return to this in Section 6.

3.4 Evaluation protocol and pre-registered kill switches

Four quantities are evaluated, always in teacher (or common-frame) coordinates. **Field error**: normalized MSE and mean cosine similarity over probe points where $\|V_t\|$ exceeds its 10th percentile. **Basin agreement**: probes are iterated to their limiting points in both models (300–1,000 steps, clipped to a sanity box); agreement is the fraction of endpoints within a tolerance ε of the reference endpoints. Because no student can be expected to exceed the source’s own stability, we always report the **self-agreement ceiling**—the same statistic for the source against itself under small probe perturbations—and the source’s convergence fraction, flagging regimes where endpoint metrics are ill posed. **Attractor recovery**: symmetric Chamfer distance between attractor sets extracted by iterating a bank of starts and merging converged endpoints. **Functional carryover**: held-out reconstruction error, for completeness.

Each experiment fixes a falsification criterion before running. For distillation: *if output distillation or static latent matching comes within 0.05 basin agreement of the field-distilled student, the field carries no proprietary information and the method is redundant*. For composition: *if merged output distillation comes within 0.05 joint agreement of the field-union student, composition is likewise redundant*. Negative outcomes are reported as results, not filtered.

3.5 Union frames and lineages

Union frames. To compose two teachers A and B with disjoint knowledge, we construct a common frame that holds A ’s chart natively and grafts B ’s chart through an affine conjugacy $T : z_B \rightarrow z_{\text{frame}}$. The union field is a gated superposition,

$$V_{\cup}(z) = \begin{cases} V_A(z) & g(z) = A, \\ T(f_B(T^{-1}(z))) - z & g(z) = B, \end{cases} \quad (4)$$

with g a nearest-centroid gate over the frame positions of the two knowledge sets. The union *oracle* iterates this gated composite map; students distill from $V_{\cup}$ data-free via Eq. (2). Two constructions of T are compared: *fitted* (least squares onto where A ’s encoder places B ’s inputs—input-consistent, but dependent on A ’s out-of-distribution behavior) and *isometric* (a pure translation placing B ’s chart, undistorted, into empty frame territory—the frame as a designed object owned by no ancestor).

Lineages. Generation 1 distills from $V_{\cup}$; generation $k+1$ distills, data-free, from generation k ’s own field. Every generation is evaluated against the original generation-0 oracle. A *contaminated* variant replaces teacher B with a memorization-regime teacher B_{mem} (trained without denoising; fragmented, weakly convergent field) to test whether the channel is selective.

Table 1: Law 1 (controlled 2D study): mean over three student seeds. Basin tolerance $\varepsilon = 0.3$; ceiling 0.99; teacher convergence fraction 1.00. Data-free arms are not expected to reconstruct (unconstrained data-space chart).

Arm	Basin agr.	Field cos.	Field NMSE	Attractor Chamfer	Recon.
A1 field (data-free)	0.86	1.00	0.0003	0.27	0.57
A2 fieldk (data-free)	0.72	1.00	0.0012	0.36	0.57
A+ fieldplus (5% anchor)	0.65	1.00	0.0012	0.41	0.0067
B0 scratch	0.00	0.04	1.01	2.08	0.0000
B1 output distillation	0.53	0.92	0.22	0.49	0.0003
B2 latent matching	0.15	0.19	1.09	0.64	0.0000

4 Experiments

All experiments use MLX on a single Apple laptop; the full suite (both 2D studies, both sweeps, the MNIST tier, and the composition/lineage study) completes in roughly twenty-two minutes of wall-clock time ($\approx 1,300$ seconds summed over all stages). Teachers are trained with a fixed seed; student seed counts are stated per experiment, and tables report means over student seeds. Code, JSON run records for every number reported, and the scripts generating every figure are released with the paper.

4.1 Law 1: the field is a sufficient training signal

Setup. The world is a mixture of four Gaussian blobs on a ring in $\mathbb{R}^2$, lifted by a fixed non-linear map to $\mathbb{R}^{16}$ (2,000 training and 400 held-out points). All models are MLP autoencoders $16 \rightarrow 64 \rightarrow 64 \rightarrow 2 \rightarrow 64 \rightarrow 64 \rightarrow 16$ with GELU activations; the two-dimensional latent space permits dense evaluation on a 40×40 grid. The teacher is a denoising autoencoder (input noise 0.1): seven attractors, convergence fraction 1.00, self-agreement ceiling 0.99. Six student arms, 4,000 steps, three seeds:

Arm	Objective	Sees data?
A1 field	Eq. (2), $z \sim \text{Uniform}(\text{box})$	no
A2 fieldk	k -step matching, $K = \{1, 2, 4, 8, 16\}$	no
A+ fieldplus	A2 + 0.2· reconstruction on a 5% anchor	5%
B0 scratch	reconstruction	yes
B1 outdistill	$\ s(x) - t(x)\ ^2$	yes
B2 latmatch	$\ E_s(x) - E_t(x)\ ^2 + \text{reconstruction}$	yes

Results. Table 1 and Figure 1. The data-free field-distilled student matches the teacher’s field almost exactly (cosine 1.00, NMSE 3×10^{-4}) and reproduces emergent quantities it was never directly supervised on: attractors coincide with the teacher’s (Chamfer 0.27 against a merge radius of 0.15) and 86% of random probes iterate to the teacher’s endpoint. The kill switch does not fire. Training from scratch on the same data yields an unrelated field (0.00)—the field is a property of the individual trained network, not of the task. Static latent matching transfers almost none of the off-manifold field (0.15). Output distillation is the strongest conventional baseline (0.53)—cloning the composite map’s behavior on data transfers part of the dynamics, whereas matching static features does not—but it falls 0.33 short of explicit field matching, with more than double the seed-to-seed standard deviation (0.17 versus 0.07).

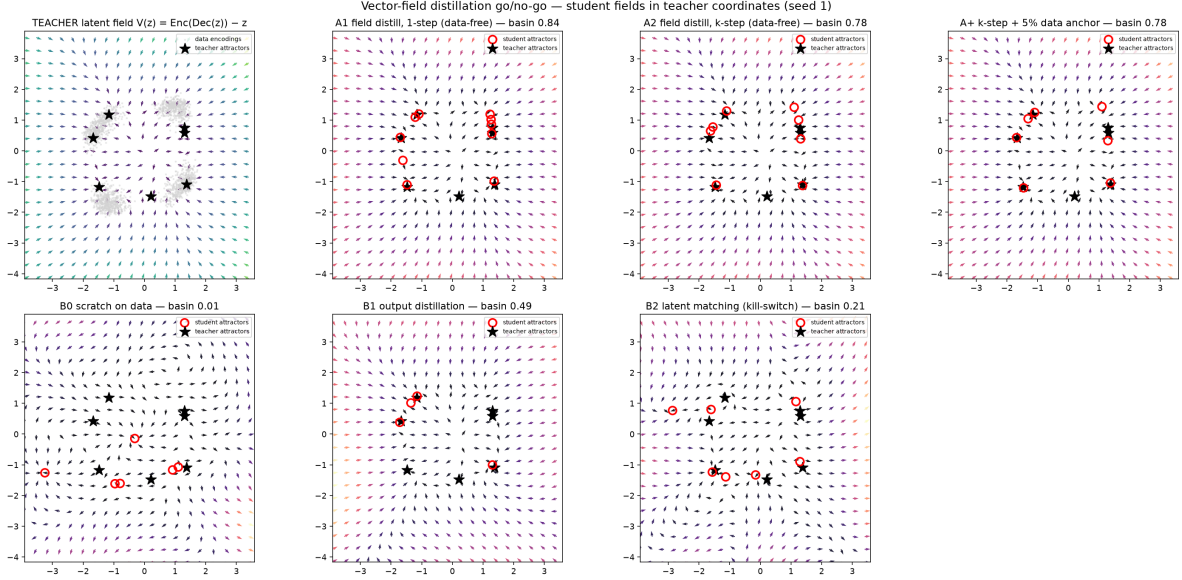


Figure 1: Teacher field and student fields in teacher coordinates (seed 1). Black stars: teacher attractors; red circles: student attractors. Field-distilled students reproduce the field and place attractors on the teacher’s; the scratch baseline bears no relation to the teacher; output distillation is partial; latent matching scatters.

The teacher’s regime is a prerequisite. An earlier run of the identical protocol with a non-denoising teacher (memorization regime: 26 fragmented attractors; only 75% of probes converge) produced basin agreement of only 0.26 for the field arm *despite* field cosine 0.97: per-step errors compound along slow manifolds, and endpoint metrics additionally become ill posed. This motivated the regime study below. The multi-step objective (A2), constructed to counteract exactly this compounding, did not help in either regime (0.72 versus 0.86 contractive; 0.13 versus 0.26 memorizing): per-step precision, not horizon supervision, is the binding constraint.

4.2 Law 2: copyability across training regimes

Repeating the field-distillation arm against teachers trained at denoising levels $\sigma \in [0, 0.2]$ (Table 2, Figure 2) reveals two phases. For $\sigma \leq 0.02$ (memorization regime) copyability is low *and unstable*: means 0.34–0.57, seed standard deviations up to ± 0.25 . For $\sigma \geq 0.05$ (contractive regime) it rises monotonically and saturates at 0.99 with zero observed variance. Low-mean/high-variance on one side, saturated/deterministic on the other: the signature of an order parameter across a phase boundary. Our stronger prior hypothesis—copyability tracks generalization as measured by off-manifold robustness—is **falsified** by the same data: robustness is optimal near $\sigma \approx 0.08$ –0.10 and *degrades* at $\sigma = 0.2$ while copyability continues to rise (Figure 2, right). Copyability tracks the *dynamical order* of the source, not its test metrics.

4.3 Law 2, localized: reliability is the discontinuous quantity

A dense sweep ($\sigma \in \{0, 0.01, 0.02, 0.025, 0.03, 0.035, 0.04, 0.045, 0.05, 0.06, 0.08\}$, three seeds per level; Figure 3) localizes the transition. The teacher’s convergence fraction fluctuates in $[0.59, 0.78]$ for all $\sigma \leq 0.045$ and snaps to 1.00 at $\sigma = 0.05$; at exactly that level the seed spread of copyability collapses (individual seeds span 0.09–0.75 below; 0.690, 0.685, 0.685 at 0.05). The *mean* is continuous through

Table 2: Law 2 (regime sweep; two student seeds per level). “Robustness” is the teacher’s reconstruction error on noise-perturbed held-out inputs (lower is better). Copyability is the basin agreement of the data-free field-distilled student.

Teacher noise	Attractors	Conv. frac.	Ceiling	Copyability	Robustness
0.00	13	0.60	0.96	0.57 ± 0.17	0.0019
0.01	28	0.78	0.95	0.41 ± 0.09	0.0017
0.02	11	0.63	0.97	0.34 ± 0.25	0.0016
0.05	3	1.00	0.99	0.69 ± 0.00	0.0013
0.08	4	1.00	0.99	0.66 ± 0.14	0.0012
0.10	7	1.00	0.99	0.81 ± 0.03	0.0012
0.15	4	1.00	0.99	0.99 ± 0.00	0.0013
0.20	4	1.00	0.99	0.99 ± 0.00	0.0017

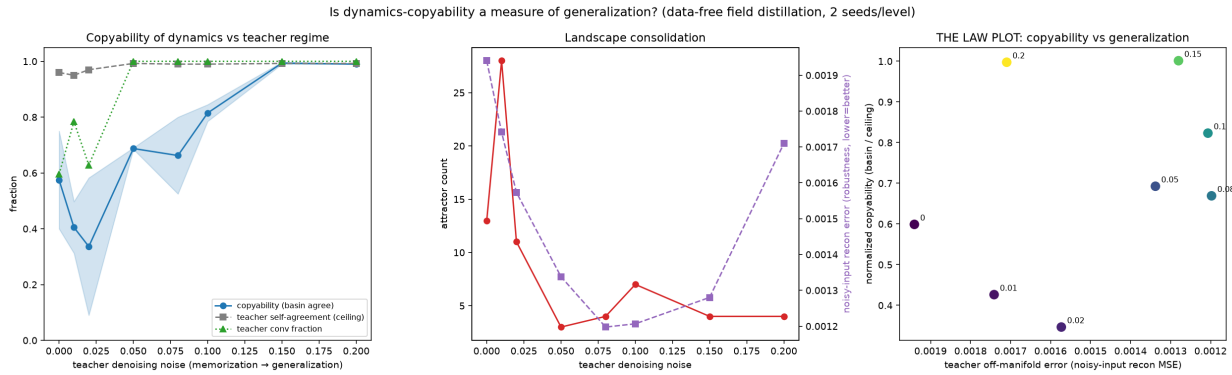


Figure 2: Copyability versus teacher regime. Left: copyability, self-agreement ceiling, and convergence fraction versus denoising noise. Center: attractor consolidation and off-manifold robustness. Right: normalized copyability against off-manifold error is not monotone, falsifying the strong “copyability equals test robustness” hypothesis.

the transition ($0.61 \rightarrow 0.69 \rightarrow 0.80$); the *reliability* is discontinuous. Transmission of dynamics becomes deterministic exactly when the source’s field becomes fully convergent. Two qualifications: attractor count is a poor observable near the transition (fluctuating between 1 and 28 with no trend), and our attempt to anchor the transition to the memorization signature of [Fumero et al. \(2025\)](#)—attractors detaching from training encodings—is confounded in this dense-data world (with 500 points per class, “on a memorized point” and “at a prototype” are geometrically indistinguishable; the statistic peaks mid-transition, 0.70 at $\sigma = 0.025$). Discriminating the two requires sparse data; we report the attempt for completeness.

4.4 Law 3, first face: the basin-level wall at 16 latent dimensions (MNIST)

Setup. Autoencoders $784 \rightarrow 256 \rightarrow 128 \rightarrow 16 \rightarrow 128 \rightarrow 256 \rightarrow 784$ on 8,000 MNIST images; denoising teacher ($\sigma = 0.3$, selected from $\{0.2, 0.3, 0.4, 0.5\}$). At 16 latent dimensions the field cannot be evaluated on a grid; field arms sample z from a Gaussian fitted to the teacher’s latent moments (a 32-number summary—a disclosed, calibration-level use of data statistics) mixed with teacher-trajectory points; 12,000 steps with cosine decay. Endpoints were additionally compared through the teacher’s decoder in pixel space; the decoded metric agreed with the latent metric everywhere.

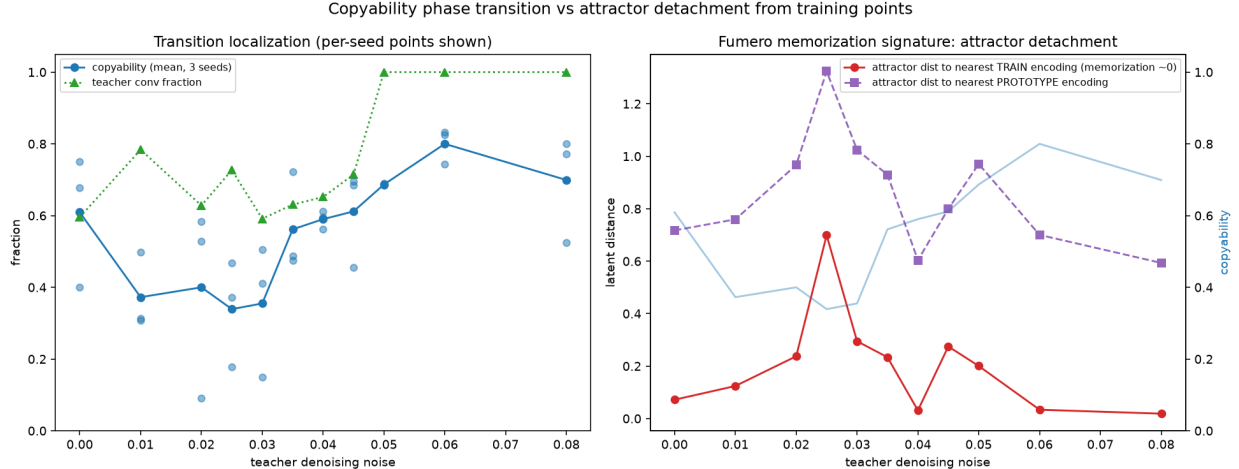


Figure 3: Dense sweep across the transition (three student seeds per level; individual seeds as scatter). Left: teacher convergence snaps from 0.71 to 1.00 between $\sigma = 0.045$ and 0.05; the spread of copyability collapses at the same level. Right: the attractor-detachment statistic is confounded in this dense-data world (see text).

Table 3: MNIST tier (latent 16; two student seeds). Teacher: denoising $\sigma = 0.3$, 18 attractors, convergence 0.97, ceiling 0.71, basin tolerance 2.19 at latent scale 5.49. “—”: no student attractor converged below tolerance. **fieldtraj** is the $K = \{1, 2, 8\}$ trajectory variant.

Arm	Basin agr.	Field cos.	Field NMSE	Chamfer / scale	Recon.
field (data-free)	0.09	0.91	0.049	—	0.106
fieldtraj (data-free)	0.01	0.90	0.068	—	0.105
fieldplus (5% anchor)	0.07	0.91	0.049	0.87	0.033
scratch	0.00	0.50	1.78	1.52	0.015
output distillation	0.19	0.70	0.67	1.08	0.016
latent matching	0.00	0.63	0.39	1.53	0.014

Results. The *ordering* of methods survives dimensionality—field distillation attains cosine 0.91 (no baseline exceeds 0.75) and the anchored variant recovers the attractor set best (Chamfer 0.87 of latent scale)—and the teacher-side phenomenon reproduces (Figure 4). But basin-level agreement fails for *every* method: the best value (0.19, output distillation) is at 27% of the 0.71 ceiling. The cause is quantified, not speculative: the field residual plateaus at NMSE = 0.049, a per-step displacement error of ≈ 0.75 latent units (14% of the latent scale)—two orders of magnitude worse, relative to scale, than the 2D study—which compounds over hundreds of iterations into median endpoint displacements of 2.5–5.8 latent units against a basin tolerance of 2.19. Three mitigations failed to close the gap: quadrupled training with decay (NMSE 0.086 $\rightarrow$ 0.049 only), trajectory-concentrated sampling, and trajectory-level supervision (**fieldtraj**, *worse*—the third failure of multi-step matching in this paper). *Superseded in v1.1:* the residual-compounding account above is accurate as a description of the parametric students but wrong as a causal diagnosis of the wall—Section 5 shows that mean residual *anti-correlates* with basin transfer across carriers of the same field, and that a lookup carrier climbs to 0.41 with no plateau as its pair budget grows. We also note that no MNIST teacher we trained exceeded a self-agreement ceiling of 0.72 at any denoising level tried (ceilings 0.51–0.72, attractor counts 18–58); v1.1 explains why: the basin boundaries are fractal ($\alpha = 0.45$, Section 5.3), so at probe scale no observer—including the teacher itself—resolves



Figure 4: Teacher attractors on MNIST decode to crisp digit prototypes (top row), reproducing the memories-as-attractors phenomenon of Fumero et al. (2025) at this scale. The anchored student’s decodes of the same latent points (bottom row) are only faintly digit-like: a 5% anchor under-constrains the student’s decoder.

Table 4: Composition (isometric union frame unless noted; oracle ceilings: joint 0.98, *A*-side 0.99, *B*-side 0.97; 9 oracle attractors, one a gate-seam artifact). Sites = oracle attractor sites recovered by the best seed.

Arm	Joint	<i>A</i> -side	<i>B</i> -side	Sites
union-field (data-free, 5 seeds)	0.82	0.90	0.75	8/9
single- <i>A</i> field (reference)	—	0.99	—	—
single- <i>B</i> conjugated field (reference)	—	—	0.81	—
merged output distillation (2 seeds)	0.13	0.27	0.00	—
union-field, <i>fitted</i> graft (hard mode, 1 seed)	0.50	0.78	0.14	5/10

them.

4.5 Laws 3–5: composition, inheritance, and selection

Setup. Eight blobs on a ring (16D lift as before). Teacher *A* is trained on blobs 0–3, teacher *B* on blobs 4–7 (denoising $\sigma = 0.1$, contractive); B_{mem} is a memorization-regime teacher (no denoising) on blobs 4–7. Union frames, gated fields, oracles, and lineages follow Section 3.5; students distill data-free for 8,000 steps. Five gates were fixed in the harness docstring before running: C1 union (joint ≥ 0.60 , per-side drop vs. single-teacher students ≤ 0.15); K3 kill switch (merged output distillation within 0.05); C3 exceedance (≥ 7 of 8 sites in one student); C4 lineage (generation-5 joint $\geq 0.85 \times$ generation-1); C2 filter (ordered graft persists; disordered graft ≤ 0.30 by generation 3 with its ordered neighbor retained). All five passed.

Composition (Law 4). Table 4. The union student holds both knowledge sets: joint 0.82 over five seeds (range 0.77–0.91; ceiling 0.98), per-side costs relative to single-teacher references of 0.10 (*A*) and 0.06 (*B*), and up to 8 of 9 oracle attractor sites in one network—no ancestor was trained on more than half the world. Merged output distillation, given *pooled data* the union student never sees, scores 0.13 with the grafted side at exactly 0.00 in both seeds: behavioral cloning has no mechanism for placing disjoint knowledge in one representation. The composition kill switch clears by 0.69.

The frame must be designed (Law 3, second face). Grafting *B*’s knowledge where *A*’s encoder places *B*’s inputs—the input-consistent choice—destroys it: the fitted-graft union student reaches only 0.14 on the grafted side (and 0.50 joint), because *A*’s out-of-distribution chart compresses foreign territory (graft-native centroid separation 0.73, versus 2.63 under the designed

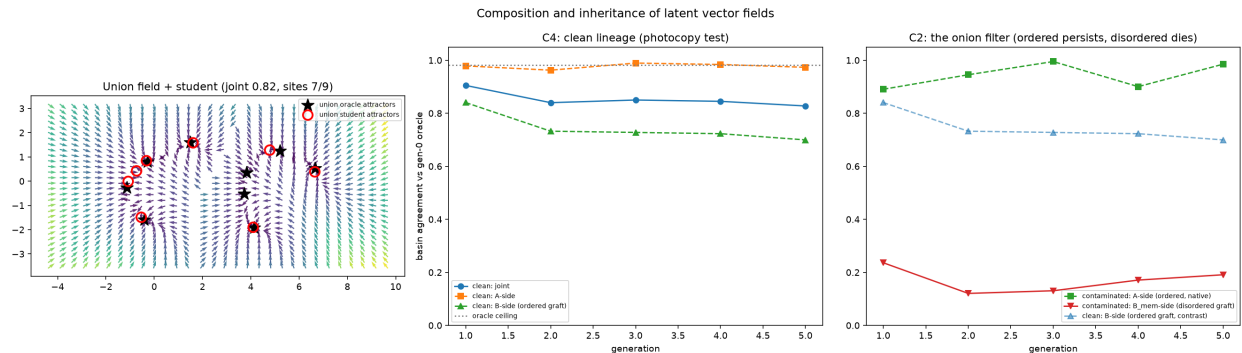


Figure 5: Composition and inheritance. Left: the union field with the student’s attractors (red) on the oracle’s (black)—one data-free student holding two disjoint teachers’ knowledge. Center: five generations of data-free re-distillation; curves are flat, not photocopy decay. Right: the selection effect—the ordered native side rides at 0.89–1.00 for five generations while the memorization-regime graft never exceeds 0.24; the clean ordered graft (blue) persists in the same slot.

frame), shrinking the local basin scale below the achievable field resolution. The identical knowledge grafted isometrically into empty frame territory transmits at 0.75–0.84. This is the same inequality that fails on MNIST, violated from the opposite side: there the residual grows; here the basin scale shrinks. Practically: a union frame is a designed object, owned by no ancestor; alignment to any single model’s chart is the wrong abstraction.

Inheritance (Law 5) and selection. Five generations of data-free re-distillation from the best union seed retain the knowledge essentially undegraded: joint $0.91 \rightarrow 0.84 \rightarrow 0.85 \rightarrow 0.84 \rightarrow 0.83$ (a 91% retention ratio against the 0.85 gate), with the native side pinned at 0.96–0.99, the ordered graft easing $0.84 \rightarrow 0.70$, and attractor counts stable at 7–10. The channel is not a photocopier. It is also selective: in the contaminated lineage, the memorization-regime graft enters at 0.24 and never recovers (0.12–0.19 across generations) while the ordered native side rides at 0.89–1.00 beside it for all five generations; the clean lineage’s ordered graft, in the same slot, persists at 0.70–0.73. Disordered structure is not merely copied poorly—it cannot board the lineage at all. (v1.1 qualification: Section 5.1 shows this exclusion is a property of the lossy parametric carrier, not of the field channel—a lossless read-only carrier transmits the same graft faithfully across all five generations.) In a preliminary half-budget run we additionally observed a *ratchet*: a weak ancestor’s grafted side improved from 0.22 to 0.71 under re-distillation before stabilizing (recorded in the released logs).

Reliability is budget-dependent. At 4,000 training steps, union-student seeds spanned 0.35–0.77; doubling to 8,000 collapsed the spread to 0.77–0.91. The union target—two structured territories and a gate seam—is a harder regression landscape, and the seam is where residual basin losses concentrate; reliability behaves as a budget knob, echoing Law 2 at the composition level.

Caveats. Lineage chains use one seed per generation, and the clean lineage’s ancestor is the best of five seeds (a disclosed selection); the clean-graft persistence criterion is met with little slack (0.70 against a 0.67 threshold); the ninth oracle attractor is a seam artifact; the hard gate discontinuity is smoothed by students. All results in this section are two-dimensional.

Table 5: v1.1 carrier swap in the composition/lineage world (identical teachers, frames, oracles and gates as Table 4).

Carrier of V_U	Union joint	Lineage $1 \rightarrow 5$	Mem. graft (B -side)
SGD student (Section 4.5)	0.82	$0.91 \rightarrow 0.83$	excluded: ≤ 0.24
read-only lattice	0.99 (ceiling 0.97)	0.99 , flat	transmitted: 0.68 , flat, 19/22 sites

5 External Validation and Re-diagnosis (v1.1)

Within a day of the v1.0 release, the second author—then an outside reader—reproduced the headline results of Sections 4.1 and 4.4 in a self-contained reimplement and filed four claims through the repository’s issue tracker:² (i) the transmission losses of Sections 4.4–4.5 belong to the SGD *carrier*, not the field *channel*; (ii) mean field NMSE anti-correlates with basin transfer—the wrong observable; (iii) the high-dimensional wall is budgetary, not dimensional; and (iv) the uncertainty exponent α of the teacher’s basin boundaries (Grebogi et al., 1983; McDonald et al., 1985) is a teacher-only order parameter that predicts the transmission ceiling. This section validates each claim inside the original harness—the same cached teachers, worlds, oracles, and evaluation code behind every number above (module `extval`; all run records released)—and reports one new experiment the exchange prompted. Claims (i), (ii) and the cost-curve half of (iii) are confirmed; the efficiency half of (iii) partially replicates; (iv) is confirmed on the original teacher (its dimension-scaling extension was reviewed against the contributed run records but not independently re-run).

5.1 The carrier, not the channel

We replace the SGD student of Section 4.5 with the simplest carrier that can hold a two-dimensional field: a read-only 512^2 lattice snapshot of the union field, queried by nearest cell. No optimization occurs anywhere in the pipeline.

Table 5: every loss that v1.0 attributed to the channel disappears. The union is held at the oracle’s own ceiling; five generations of re-transmission are flat rather than decaying; and—decisively—the memorization-regime graft, which the SGD lineage excludes, rides the carrier lineage indefinitely. Law 5 is therefore restated: the field channel is nearly lossless and *unselective*; both the heredity losses and the selection effect of Section 4.5 live in the lossy parametric carrier at fixed budget. Selection is not an intrinsic virtue of the channel but a *designable property of the carrier*: choose a budget, choose what cannot board.

5.2 The wrong observable

On the MNIST teacher of Section 4.4 we tabulated 10^6 pairs $(z, V_t(z))$ from the same sampling prior used to train the field students, and compared carriers of the identical field under the evaluation of Section 3.4. A nearest-neighbor lookup over the full table has mean NMSE 0.397—eight times *worse* than the parametric student’s 0.049—and basin agreement 0.410, $4.6\times$ *better* than the student’s 0.090. Sharper still: selecting the kernel width of the Gaussian-process reconstruction below by held-out NMSE yields basin agreement 0.000–0.004 in all three seeds, while selecting by directional (cosine) preservation yields partial transfer. Mean residual magnitude is the wrong observable; basins are decided by where the displacement *points* in boundary-adjacent—decision-relevant—regions, not by its average size. The ρ (residual-to-basin-scale) inequality of v1.0 is withdrawn as

²<https://github.com/Veso-AI-Open-Source/knowledge-as-dynamics/issues/1>, July 9–10, 2026.

Table 6: v1.1 carriers of the MNIST teacher’s field (three seeds; evaluation of Section 3.4; ceiling 0.71). References: parametric field student 0.090, output distillation 0.191 (Table 3).

Carrier	Teacher queries	Basin agreement
k -NN lookup	16,384	0.02–0.06
k -NN lookup	262,144	0.27–0.31
k -NN lookup	10^6	0.410
GP posterior mean, σ by NMSE rule	16,384	0.000–0.004
GP posterior mean, σ by cosine rule	16,384	0.095 ± 0.060

stated; its geometric core survives in the chart-compression failure of Section 4.5, where a foreign encoder’s chart collapses the margins *between* basins.

5.3 The wall is budgetary; the ceiling is fractal

Table 6: lookup basin agreement climbs $0.04 \rightarrow 0.28 \rightarrow 0.41$ over three decades of budget with no plateau, already $4.6\times$ the best v1.0 parametric arm at 10^6 pairs. The MNIST wall is a sample/compute budget, not a property of dimension.

The ceiling is a different object. Following Grebogi et al. (1983), we measured the teacher’s final-state sensitivity: the fraction $f(\varepsilon)$ of ε -perturbed probes whose endpoint changes scales as $f(\varepsilon) \propto \varepsilon^\alpha$, with α the uncertainty exponent and $D_b = d - \alpha$ the basin-boundary dimension (McDonald et al., 1985). On the cached v1.0 teacher: $\alpha = 0.454$ with convergence fraction 0.97—boundaries of dimension ≈ 15.5 in a 16-dimensional latent space, nearly space-filling, despite fully convergent dynamics. The exponent *predicts the ceiling*: at the evaluator’s probe perturbation, $1 - f(\varepsilon_{\text{probe}}) = 0.69$ against the measured self-agreement ceiling of 0.711. Two consequences. First, the ≤ 0.72 ceilings of Section 4.4 are the teacher’s own boundary geometry, not an evaluation artifact. Second, convergence fraction is an insufficient certificate of dynamical order (0.97 here, with $\alpha = 0.45$): Law 2 acquires a sharper, teacher-only diagnostic that can be measured *before* any transfer is attempted.

5.4 Non-parametric efficiency: a partial replication

The external report’s strongest quantitative claim was that a Gaussian-process posterior-mean reconstruction of the field from only 16,384 pairs (RBF kernel ridge; the Wiener-filter lineage of Hoffman and Ribak, 1991; see Rasmussen and Williams, 2006) reaches basin agreement 0.48—what lookup needs $\sim 10^6$ pairs for—on a teacher retrained from the released recipe. On the *original* cached teacher this does not reproduce: the best GP configuration (cosine-rule σ , an interior maximum of a grid spanning $0.05\text{--}0.5\times$ the median pairwise distance) attains 0.164/0.102/0.020 over three seeds (0.095 ± 0.060 ; Table 6). The reimplementations’ sampling procedure is equivalent to ours, so the divergence is teacher-level, and it runs in both directions—the external lookup at 10^6 pairs is *worse* (0.32) than ours (0.410) while its GP is far better—consistent with correlation-structure gains that depend on the smoothness of the particular teacher’s field. We record this as an open, teacher-dependent question rather than a refutation.

5.5 Two-hop distillation: the parametric bottleneck is reliability

The carriers above hold the field without compression; the v1.0 program, however, needs *parametric* students. The exchange prompted a new experiment: supervise a standard parametric student not by querying the teacher in Eq. (2) but by querying a cheap non-parametric reconstruction—the

16,384-pair GP—as a dense oracle, with architecture, step count and sample budget identical to the field arm of Section 4.4.

Across six seeds the result is bimodal: basin agreement 0.008/0.387/0.004/0.309/0.496/0.004—half the seeds fail outright, half land at 0.31–0.50 (mean 0.40). The successes are notable on three counts. The best student (0.496) exceeds the 10^6 -pair lookup (0.410) while consuming $61\times$ fewer teacher queries, reaching 70% of the ceiling where the direct field arm reached 13%. Students exceed their own supervisors: the GP reconstructions score 0.02–0.164 used directly as carriers, and the seed-2 student reached 0.387 against its own supervisor’s 0.020—distillation *through* a low-basin carrier can land far above it. And the failure mode is invisible in field space: all six students sit at cosine 0.82–0.83 and NMSE 0.55–0.58; only basin agreement separates the modes—the wrong-observable phenomenon of Section 5.2 recurring one level up, at the supervision signal. The parametric wall of Section 4.4 is therefore a *reliability* problem, not a capacity problem, echoing Law 2’s discontinuity in yet another guise: what is discontinuous across seeds is whether transmission happens at all.

6 Discussion

One geometry explains both failures (revised in v1.1). v1.0 unified this paper’s two negative results under a single inequality: defining ρ as the ratio of achievable per-step field residual to local basin scale, faithful transmission requires $\rho \ll 1$ —with MNIST inflating the numerator and the fitted graft shrinking the denominator. The external validation of Section 5 falsifies that unification as stated: carriers with far larger mean residual transfer basins far better, so the numerator was the wrong quantity. What survives is the geometric core, now stated with the right observables. Transmission fidelity is governed by *directional fidelity in decision-relevant regions*—probes near basin boundaries—relative to the geometry of those boundaries: their measure (the uncertainty exponent α sets the ceiling any carrier can reach) and their margins (chart compression in the fitted graft collapses inter-basin separation by $3.6\times$, destroying the decision-relevant geometry itself). On MNIST the boundaries are near-space-filling ($D_b \approx 15.5$) and the binding constraint is budget; in the fitted graft the boundaries are annihilated and no budget recovers them. A framing that accounts for its own failures—and for its own mistaken first diagnosis—with the machinery of its successes is, we believe, stronger evidence for the field-as-channel abstraction than the v1.0 claim was.

What the results establish. Within the controlled setting: the latent vector field is a sufficient, data-free carrier of a network’s learned structure, and this information is demonstrably absent from the standard distillation channels (Table 1); its transmissibility is an intrinsic property of the source with a sharp reliability transition at the source’s dynamical ordering (Table 2); fields from disjoint sources compose in one student under a designed frame, which no behavioral channel achieves (Table 4); and the composed knowledge survives generational re-transmission while memorization-regime structure is excluded from the lineage (Figure 5), an exclusion that v1.1 locates in the parametric carrier rather than the channel (Table 5). Table 7 states each law at its demonstrated scope alongside its falsifiable general form.

What they do not establish. All positive results are at laptop scale with two-dimensional latent spaces where the field can be densely sampled; at 16 latent dimensions parametric students do not deliver basin-level fidelity directly—v1.1 reframes that wall as carrier budget plus seed reliability rather than a barrier of dimension, but crossing it cheaply *and dependably* with a parametric carrier

Table 7: The five laws: demonstrated scope versus conjectured scope. Every conjecture is falsifiable with the released harness plus scale.

Law	Demonstrated (this paper)	Conjectured scope
1. Sufficiency	2D AE: 0.86 vs. 0.53/0.15 baselines, data-free	any re-encoding system with a convergent field; knowledge transfer where data cannot move
2. Order	reliability discontinuous at full convergence ($\pm 0.25 \rightarrow \pm 0.00$ at $\sigma \approx 0.05$); $\alpha = 0.45$ predicts the MNIST ceiling (0.69 vs. 0.71) where conv. fraction 0.97 does not	a data-free diagnostic of the memorization-generalization boundary, with α as its teacher-only certificate
3. Resolution (v1.1)	mean NMSE anti-correlates with basin transfer; lookup carrier climbs without plateau (0.41 at 10^6 pairs); chart compression collapses inter-basin margins (fitted graft 0.14)	directional fidelity near basin boundaries as the governing constraint for scaling and for frame design
4. Composition	union 0.82 vs. behavioral merging 0.13	multi-model merging via designed (ultimately nonlinear, learned) frames
5. Heredity + selection (v1.1)	SGD lineage: 91% retention, memorized structure excluded; lattice carrier: flat at ceiling, memorized structure transmitted	selection strength as a <i>designable</i> function of carrier budget in lineages that accumulate general structure

remains undemonstrated, and every operation in Section 4.5 inherits that caveat. Nothing here concerns reasoning, agency, or capability gain: the union student knows the union of its ancestors’ knowledge and nothing more; the lineage accumulates and purifies but does not discover. The falsified robustness hypothesis cautions against reading copyability as a general measure of model quality: it measures dynamical order. “Inheritance” and “selection” are used in the operational senses defined by the gates of Section 4.5.

Methodological limitations. Seed counts are small throughout (two to five student seeds; one teacher seed per condition; one seed per lineage generation, with a disclosed best-of-five ancestor selection). The alignment protocol is affine-only and may under-credit baselines whose charts differ nonlinearly from the teacher’s, though it is applied best-case for every arm. Endpoint-based basin metrics are partially ill posed when the source’s field does not fully converge; convergence fractions and ceilings are reported throughout. The attractor-detachment analysis is confounded by data density and inconclusive as reported. The MNIST sampling prior uses the teacher’s latent moments (32 numbers, disclosed). The union gate is a hard partition and its seam concentrates the residual losses.

Outlook. The five laws define a program whose blocker v1.1 has sharpened: not driving mean residual down, but delivering directional fidelity near basin boundaries *reliably* through a parametric carrier. Candidate attacks follow directly from Section 5: boundary-weighted sampling (importance-weight the decision-relevant regions, not the attractors), directional objectives in place of magnitude regression, two-hop supervision through non-parametric reconstructions with seed selection or ensembling to manage the reliability mode, α -screening of teachers before transfer, and—where a parametric form is not required—direct carrier transmission, which already crosses the wall at budget. If these hold at scale, the demonstrated operations become practical: knowledge

transfer when data cannot be shared, model merging as field union under learned nonlinear frames, and lineages that accumulate the transmissible (general) core of many specialists while shedding what was memorized at a chosen carrier budget. Nearer-term: resolving the teacher-dependence of non-parametric reconstruction efficiency (Section 5.4); sparse-data worlds to unconfound the detachment signature; teacher-seed ensembles at the transition; frame design as an optimization problem in its own right; and porting the channel machinery to systems whose fields are trained rather than emergent (Ho et al., 2020; Song et al., 2021), including recurrent-depth models whose step displacements are already measured (Pappone et al., 2025).

7 Conclusion

We proposed treating a neural network’s learned knowledge as its latent iteration dynamics, and characterized the displacement field $V(z) = E(D(z)) - z$ as a transmission channel. Five pre-registered, laptop-scale experiments yield five regularities: the field alone suffices to transfer an attractor landscape without data; transmissibility is an order parameter of the source, with reliability changing discontinuously at full convergence; basin-level fidelity is governed not by mean field error—an observable that anti-correlates with transfer—but by directional fidelity near basin boundaries, whose fractal dimension sets every carrier’s ceiling; fields from disjoint specialists compose in a designed common frame, where behavioral merging fails entirely; and the composed knowledge is heritable across generations of data-free re-distillation, with the selection against memorized structure located in the carrier, not the channel. An independent replication arrived within a day of release; its re-diagnosis is validated in Section 5 and incorporated throughout—exactly the correction cycle this paper’s release model was designed to invite. The latent vector field, so far an analysis tool, is a channel: it can teach, it can merge, and it can inherit; read-out carriers already cross the high-dimensional wall at sufficient budget, and two-hop distillation shows parametric students can follow. The gate between these demonstrations and practice is now twofold and well-specified: directional accuracy in decision-relevant regions at scale, and the reliability of the parametric carrier.

Contribution statement

E.H. conceived and built the v1.0 harness and experiments, wrote v1.0, ran the v1.1 validation experiments, and wrote the present text. I.N. entered the project one day after the v1.0 release with an independent, from-scratch reimplementation of the harness, a replication of its headline results, and the four-part re-diagnosis filed as repository issue #1 and validated in Section 5: that the transmission losses and the selection effect belong to the parametric carrier rather than the field channel, that mean field error is the wrong observable for basin transfer, that the high-dimensional wall is budgetary, and that the uncertainty exponent α is the right teacher-side order diagnostic. Laws 3 and 5 are restated, and Law 2 sharpened, on the strength of those contributions; the two-hop experiment of Section 5.5 arose from the exchange. I.N. joined as co-author at v1.2.

Reproducibility

All experiments were implemented in MLX and run on a single Apple M-series laptop; the complete suite (controlled study, both sweeps, the MNIST tier, and the composition/lineage study) executes in roughly twenty-two minutes of wall-clock time ($\approx 1,300$ seconds summed over all stages). The released harness contains one module per experiment with fixed seeds, JSON run records for every number reported here, and the scripts that generate every figure: <https://github.com/>

[Veso-AI-Open-Source/knowledge-as-dynamics](#). The v1.1 external-validation module (`extval`: the carrier swap, the 10^6 -pair table and carrier cost curve, the α measurement, and two-hop distillation) runs in roughly another twenty-five minutes; its JSON run records are released alongside the originals.

Revision history

v1.0 (July 9, 2026): initial release. **v1.1** (July 10, 2026): added Section 5 (external validation: carrier swap, carrier cost curve, uncertainty-exponent measurement, two-hop distillation); restated Law 3 (mean-residual diagnosis withdrawn in favor of directional fidelity near basin boundaries) and Law 5 (selection located in the carrier); sharpened Law 2 with the uncertainty exponent α ; revised the Discussion and Conclusion accordingly; prompted by repository issue #1. **v1.2** (July 10, 2026): Ivan Nemytchenko joins as co-author; contribution statement added; results unchanged.

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